

Using Messy Text in Future LLM Annotations*

Patrick Brandt¹ Shreyas Meher² Paula Cuellar-Cuellar¹ Xingyuan Zhao¹

¹The University of Texas at Dallas

²Erasmus University Rotterdam

May 6, 2026

Abstract

When working with real-world text, researchers often inherit corpora and annotations with costly human judgments but were not collected, organized, or annotated for modern text-as-data workflows. This is especially common in social science domains where web-scraped documents are noisy, incomplete, or misaligned with the unit of analysis. We propose *codebook attention*: a codebook-guided turn-by-turn extractive summarization pipeline that reuses valuable annotations without requiring full re-annotation. For each victim in a corpus of disappearance reports from Mexico, the model processes documents sequentially, extracts verbatim evidence spans for each codebook category, and updates a running structured summary to produce a clean and evidence-traceable document-level representation. Using the codebook as the task schema focuses model attention on the fields that matter for downstream classification and reconnects prior human annotations to contemporary language model workflows. We evaluate the resulting summaries by classifying and comparing the predicted labels with human annotations across multiple open-source models including Llama 3.1, Gemma 3 and Ministral 3. Results suggest that hours of computation can approximate weeks of human coding effort while producing evidence-traceable summaries suitable for downstream human rights and event data research.

*This research was supported by NSF award OAC-2311142 and used Delta at NCSA / University of Illinois through allocation CIS220162 from the Advanced Cyberinfrastructure Coordination Ecosystem: Services & Support (ACCESS) program, which is supported by NSF awards 2138259, 2138286, 2138307, 2137603, and 2138296. Any opinions, findings, and conclusions or recommendations expressed in this material are those of the author(s) and do not necessarily reflect the views of the National Science Foundation, NCSA, or the affiliated university resource providers.

The data here were created initially by researchers from the Human Rights Program at the University of Minnesota to explore Mexican press reporting in four states in Mexico from 2009-2018. The Press Reporting project is affiliated with the Observatory on Disappearances and Impunity in Mexico, based at UNAM and FLACSO in Mexico City (<https://odim.juridicas.unam.mx/>). We acknowledge the Principal Investigator on the initial data collection and annotation, Barbara Frey, J.D., director of the Human Rights Program from 2001-2022. The Project Coordinator was Maria Ignacia Terra. The primary researchers on the project included: Yolanda Burckhardt, Maria Larson, Rosaleen Joyce, Hunter Johnson, Olga Salazar Pozos and Paula Cuellar Cuellar. Thanks also to Dr. Carrie Booth Walling, director of the Human Rights Program, for ongoing maintenance of the database. The database was supported with funding from Fulbright-COMEXUS, the University of Minnesota Human Rights Initiative and the Ohanessian Endowment Fund for Justice and Peace Studies.

1 Introduction

Many valuable text data and annotations were created before the standards and data structures now expected in LLM based research. These materials preserve costly human judgment and domain knowledge, yet they were often not gathered, organized, or annotated in ways that make them directly usable for contemporary computational workflows. This creates a central research problem: how can researchers use or reuse legacy texts and annotations without losing their original knowledge while making them usable for modern LLM based analysis.

Our goal is to reuse the prior data collection and annotations to create a balanced and clean corpus. Addressing downstream tasks using real-world data, researchers often must work with messy texts. In this research, we refer to the text that has heterogeneous noises, confounding content, and structural misalignment as *messy texts*. These texts are scraped websites and documents from previous works that may contain irrelevant formatting and information resulting from poor web scraping, connection errors, or faulty parsing, such as navigation menus, error pages, cookie consent notices, and social media sharing buttons interspersed with article content, creating noise that will distract the model from the useful content. What might be worse is that, these texts might contain seemingly related but essentially unrelated information as distractions, originating from suggested articles or user comments on the article page, introducing hallucinations and conflation of events for the LLMs. Additionally, these texts might not be organized or structured in a way that is ready for processing by modern text-as-data methods, making it unsuitable for the LLMs for it will further confuse the model.

Our example here is based on a corpora of annotated texts about disappearances in Mexico. The initial collection of these heterogeneous corpus requires a systematization and coding, applying a consistent analytical framework to materials that were not originally collected under uniform criteria. Each case identified in the press corpus was reviewed and, where possible, translated into a structured set of variables capturing key dimensions of the disappearance. But given the nature of the original corpus, this often involved interpretive reconstruction. Press reports often contain incomplete, inconsistent, or ambiguous information, requiring careful reading and, in some cases, comparison across multiple articles related to the same case. Even when multiple reports exist, discrepancies between sources are common, particularly in relation to causes of disappearance, involvement of authorities, or outcomes of investigations.

The press-based corpus analyzed in this study originates not as a fully standardized database but as a fragmented and heterogeneous collection of newspaper articles. Through processes of systematization and coding, this study transforms these materials into a structured analytical dataset that enables the examination of both patterns of disappearance and the ways in which they are represented in public discourse. The limitations and inconsistencies of the corpus are integral to the analysis, revealing the challenges of documenting and understanding disappearances in contexts marked by violence, uncertainty, and impunity.

The original annotations were made with the victim as the unit of analysis, recording codebook fields such as method of capture and who carried out the threats for each person. As for the corpora

itself, the documents were collected and organized by news source rather than by victim, so any single victim’s record is distributed across multiple documents whose linkage back to the annotations cannot realistically be reconstructed by human labor. This misalignment matters for downstream use, since fine-tuning and classification over news reports operate on individual documents and therefore require labels that hold at the document level. A victim’s method of capture may appear in one report while the identity of those who carried out the threats appears in another, yet both were annotated at the victim level without reference to a specific document. Propagating victim-level annotations onto every document in which the victim appears would therefore train downstream models on incorrect signals, because many of those documents do not contain the evidence for the label they would inherit.

Humans can easily filter and extract around this noise and may have already done so to produce annotations. But to extend this with the latest LLMs one needs to a) clean the texts (explicitly) and b) reconnect the human annotations to the modern needs to LLMs (e.g., span annotations rather than just words or linkages to actual images in a larger document) (Olsen et al. 2024). The core problem here is therefore not only cleaning noisy text, but also aligning annotations created by human coders who can infer structure from dispersed evidence across multiple documents, with the exact source spans where these annotations come from, using LLM workflows.

Solving these problems allows researchers to work with and re-use texts for their research but that are currently messy and not well-structured. For example, legacy annotated corpora may suffer from that the codings are not linked to the spans or chunks in a level that is suitable for modern text as data research (Escott 2018). Corpora that has multi-document aligned with other units of analysis may suffer from that evidence about a single case is dispersed across thousands of pages and many persons, making the material hard to navigate and requires additional retrieval structures that can be solved by using LLMs (Gibbs et al. 2010). And digitized archives and other scraped web corpora such as the congressional hearings or news archives also face semantic confounding contents that cannot be simply removed by rule-based approaches (Hurtado Bodell et al. 2022; Molina et al. 2025). Our application is a corpus on disappearance cases that is valuable and significant for human rights’ research as well as serving training for the classification and prediction of future abduction cases. We propose a multi-turn extractive summarization method that sequentially processes the original messy and disorganized documents for each unit of analysis (victim). This is done with a conversation state that iteratively updates the running summary with exact text spans from new documents, generating a summary by item for each victim, comprised of sentences extracted from the report that are directly relevant to the item.

From a broader point of view, this work contributes a generalizable pipeline for aligning prior victim-level annotations with document-level source spans, producing evidence-traceable outputs that reuse existing codebooks without re-collection. This work also demonstrates a better way of using LLMs to assist social science research that is more robust to hallucination and other biases by using a multi-turn stateful workflow framework that maintains context across operations and allows tracking the source and evidence of the information extracted from the text.

A further contribution concerns the use of the pipeline output as training data for downstream model adaptation. The codebook-guided summaries produced by this workflow are structured, label-aligned, and free of the noise that makes the original scraped documents unsuitable for supervised learning. Each summary is organized by codebook field, with extracted evidence spans linked to their source documents, which means the pipeline output can serve directly as input–output pairs for fine-tuning smaller, task-specific classifiers.

2 Related Work

We review and consider three approaches for aligning and re-using our prior texts and their annotations. These are detailed below as *non-LLM methods*, and *LLM-methods* both of which are mainly extractive approaches that would either use human or text-as-data methods. The final approach we advocate and develop is a hybrid approach that uses both generative and extractive LLM methods, which we call *extraction and summarization* (see Brandt et al. 2025 for a discussion of these distinctions).

2.1 Non-LLM Methods for Messy Text

Producing a clean, structured dataset from noisy web-scraped text while preserving traceable connections to the original source material has typically relied on rule-based preprocessing, manual data cleaning, or a combination of both (Bohannon et al. 2007; OpenRefine Community 2025). Rule-based methods can remove structural elements, but because each source website uses a different page layout they do not generalize reliably across heterogeneous sources and cannot handle semantically related but contextually irrelevant content such as suggested article previews or reader comment threads (Rahm 2011; C. Ma et al. 2025). Human labor can handle these contextual distinctions, but at the volume of a large web-scraped corpus it becomes prohibitively slow, expensive, and inconsistent, making both approaches insufficient for the noisy corpus examined in this paper (C. Ma et al. 2025; Ni et al. 2024).

Both rule-based preprocessing and human labor are insufficient for handling the noisy corpus examined in this paper, as the volume, heterogeneity, and semantic complexity exceed what non-LLM methods can realistically manage.

2.2 LLM Methods for Messy Text

Recent work suggests that LLMs are well suited to messy text beyond the limits of rule based methods because they can use contextual reasoning to distinguish relevant content from noise across heterogeneous sources. These literature include both direct cleaning tasks and related annotation tasks, together showing that LLMs can support data preparation when rule based methods are too rigid for noisy text.

LLM based work has been applied to messy news text. For example, Multi-News+ uses LLMs to classify noisy news reports into categories, which makes it similar to our task in the basic sense that both use LLMs to separate useful text from noise (Choi et al. 2024). It does not directly solve our problem, however, because our goal is not to classify already defined document inputs, but to identify and preserve relevant information within noisy documents while keeping it traceable to the source text. This work contributes by showing that LLMs can handle messy text and support data quality tasks. A another closely related project is the application Ventana a la Verdad, a retrieval-augmented chatbot over the Colombian Truth Commission’s archives, which targets interactive question answering rather than codebook-aligned summaries linked to document-level source spans (Sokol et al. 2025).

LLMs can also perform the data standardization and data error processing tasks that rule based preprocessing was originally meant to provide, because they can infer these transformations from context rather than relying on site specific rules (L. Ma et al. 2023; H. Zhang et al. 2024; Choi et al. 2024; Biester et al. 2024; Li et al. 2024; Yan et al. 2024). LLMs can also perform data integration, especially entity matching, and this supports annotation source tracing by preserving the link between each extracted annotation and the source document or spans from which it originates (Peeters et al. 2025; Wang et al. 2025; Fan et al. 2024). By contrast, data imputation does not fit our purpose because our goal is not to infer absent information, but to identify and preserve information that is already present in noisy text, and generating plausible missing values would weaken traceability to the source text and increase the risk of unsupported outputs (Ding et al. 2024; He et al. 2025). These existing works suggest that LLMs are capable of distinguishing relevant content from noise and of addressing these data preparation problems in heterogeneous text, providing the foundation for the design developed in the following sections (Zhou et al. 2026).

2.3 Extraction and Summarization

The literature that does summarization using LLMs usually assumes cleaner and more standardized inputs than the corpus examined in this paper. Many methods operate on texts that already have stable document boundaries (Choi et al. 2024), clearer annotation targets (L. Ma et al. 2023), or outputs defined at the same level as the input text (H. Zhang et al. 2024). Our setting differs because the corpus is messy, the source documents were not aligned with the unit of analysis for the downstream tasks, which creates a specific methodological choice for the paper. Pure cleaning is not enough because it does not recover the missing link between earlier annotations and the source text, and unconstrained generation is also not enough because paraphrase can weaken grounding and traceability (Brandt et al. 2025). The task instead requires a hybrid design that extracts source grounded evidence from messy documents (Dagdelen et al. 2024) and uses that evidence to produce clean structured summaries for downstream use (Olsen et al. 2024), so that they will produced in the form that current extraction and summarization workflows require while preserving human annotations. Reannotation would require time, funding, and domain expertise that are no longer available at the necessary scale, while leaving the legacy annotations untouched would preserve the

same gap between old coding work and modern LLM inputs.

The methodological problem is therefore to design a workflow that recovers traceable evidence from messy text, produces clean summaries for downstream use, and reuses prior annotations without pretending that the original human process can be recreated. Even though the related work has already shown LLMs are suitable for this task, but the others approaches cannot be simply reapplied here. We therefore choose a combined approach that uses LLMs to extract source grounded evidence and produce clean structured summaries, so that prior annotations can be reused without full reannotation and downstream tasks can operate on text that is both traceable and usable.

3 Method

To address the mismatch between victim level annotations and document level source reports while providing a clean summarization, the workflow is organized into three linked tasks. For each report, it identifies source grounded evidence, converts that evidence into a structured document level summary, and then revises the running summary as later reports add or refine information. Figure 1 illustrates this document to summary update process, and the following subsections explain each stage.

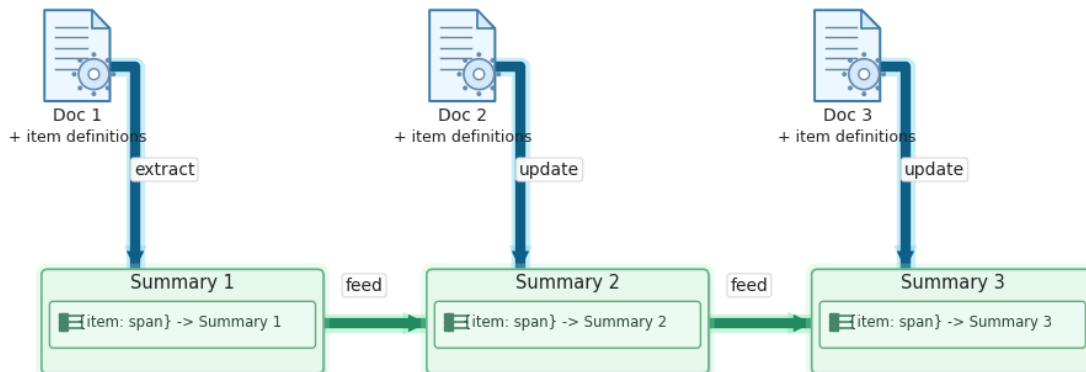


Figure 1: Workflow for extracting evidence, generating a structured summary, and updating the running summary across sequential documents.

The task is to reuse the existing codebook and human annotations by aligning them with the original texts, extracting source grounded evidence for each coded field, producing clean structured document level summaries, and updating those summaries across multiple reports while preserving traceability to the supporting spans. The following subsections explain how each of these tasks is implemented.

3.1 Task Definitions

The task is to transform a set of messy, multi document news reports associated with a single victim into a structured, evidence traceable document level representation that supports reuse of the existing codebook and annotations.

Another critical requirement is that the output text must be noise-free, so that any further LLM tasks can be meaningful. The text must combine extraction of verbatim spans with structured aggregation across documents, so that the resulting representation is both clean enough to serve as LLM input and anchored tightly enough to the source material to support reuse of the existing coded data and codebook definitions.

3.2 Model Setup

All LLM calls were executed using instruction tuned models served through vLLM (Kwon et al. 2023). The default configuration uses the AWQ INT4 quantized Meta Llama 3.1 8B Instruct model (Dubey et al. 2024), and cross model comparisons use Meta Llama 3.1 70B Instruct, Gemma 3 12B IT (Gemma Team 2025), and Ministral 3 8B Instruct (Mistral AI 2026). We use deterministic decoding with temperature 0.0. Generation is capped at 1024 tokens for summarization and 256 tokens for classification, and outputs are constrained to a fixed JSON schema to produce consistent structured fields for extracted spans and label predictions.

Table 1: Processing time per model.

Model	Time	GPU
Llama-3.1-8B-Instruct-AWQ-INT4	~5 h 27 min	A100
Ministral-3-8B-Instruct-2512	~51 min	A100
Gemma-3-12B-IT-INT4-AWQ	~38 min	A100

The Llama-8B server is the only one launched with `-quantization awq`, which forces vLLM’s legacy AWQ kernel, a kernel whose $4\times$ INT4 speedup is achieved only at batch size 1 and deteriorates rapidly as requests are batched together (Frantar et al. 2024), while Gemma’s auto-selected `awq_marlin` and Mistral’s native FP8 path both stay near peak throughput at our concurrency.

3.3 Pipeline

The pipeline takes each victim’s documents one at a time in sequence, extracting relevant spans, generating a label-structured summary anchored to those spans, and updating the summaries using previous summaries as context. This pipeline satisfies the task definition by producing document level outputs that are clean enough for downstream language model use while keeping each coded field traceable to evidence in the original documents.

3.3.1 Extraction of evidence

In the first step of the pipeline, we extract evidence rather than only generating a paraphrased summary. For each input document, the model is instructed to return verbatim text spans for each item in the codebook schema, producing a structured evidence record keyed by label. This creates traceability because every coded field is paired with the exact source phrase that supports it, so downstream users can verify where the information came from in the original report. The evidence output is designed to demonstrate that the link between each annotation and its source text is preserved and can later be verified, and it also includes a character index that locates each span within the document text. This index records how many characters into the document the extracted span begins, so a future researcher can find the exact spot in the original text and confirm the evidence.

3.3.2 Summarization with Context and Evidence

In the second step of the workflow, we generate a coherent summary while keeping it anchored to evidence and guided by the codebook schema. The model is given the document text together with the label definitions that specify the meaning of the labels and how the human coders would code the information, and it is also provided with the extracted span structure so that each sentence or clause of the summary can be linked back to the corresponding evidence. For example, the first sentence may be about the victim, and the second sentence may be about the method of capture, and the third sentence may be about who carried out the threats, instead of condensing the information into a single sentence where all the information is mixed together. This makes the summary different from vanilla summarization because it is not an unconstrained paraphrase of the document, rather, with the paragraph structured label by label, it will be easier for an LLM to pick up the right information. At this stage, summarization is performed at the level of a single document and is designed to be clean, structured, and directly usable for downstream models.

3.3.3 Updating Summary with state

In a typical single request, an LLM is stateless, meaning that it does not retain information from earlier calls, but our workflow preserves memory by maintaining an explicit conversation state for each victim as their documents are processed. The workflow maintains an explicit conversation state for each victim that accumulates across documents as they are processed sequentially. The prompt instructs the model to check each new document against the existing summary and update only the fields where new information is found. If a document contains no relevant information, the runner keeps the previous summary unchanged rather than replacing it with the model’s output. Each turn also records the output of that turn, preserving what was extracted and what summary was produced at that step for later inspection and reuse. When a document contains no relevant information, the workflow avoids replacing the existing running summary, so the accumulated context is not lost as the sequence progresses.

4 Data

4.1 Press-based Corpus Data Source

The data here are based on a press-derived corpus of disappearance cases in Mexico, developed in collaboration with the University of Minnesota’s Human Rights Program, Oxford University, Universidad Autónoma de México (UNAM), and the Facultad Latinoamericana de Ciencias Sociales (FLACSO-México), as part of broader research on enforced disappearances and media representation. The corpus consists of newspaper articles collected over time, primarily from regional outlets, with a focus on Nuevo León, Coahuila, Tamaulipas, Jalisco, and Guerrero. which have experienced sustained patterns of disappearances since the escalation of violence after 2006.

Rather than originating as a unified or systematically constructed database, the corpus was initially assembled as a collection of newspaper “clips” gathered by multiple coders and researchers working at different moments in time. These materials were identified through keyword searches using variations of terms such as *desaparición*, *secuestro*, and *levantón*, and were selected when they referred to identifiable individuals reported missing or whose whereabouts remained unknown. As a result, the corpus did not follow a single, standardized methodology at the stage of collection. Instead, it reflects an accumulative process shaped by diverse research practices and institutional contexts. Therefore, there is considerable variation in the criteria used to identify cases, the depth of information recorded, and the degree to which cases were tracked over time. Some cases are documented through multiple articles, while others appear only once, with no follow-up reporting. This lack of initial methodological uniformity is not merely a technical limitation but an important feature of the data. It reflects the broader fragmentation of information surrounding disappearances in Mexico.

Several limitations arise from the nature of the press corpus. First, the dataset is characterized by initial methodological inconsistency, reflecting its origins as a collection of articles compiled without a unified protocol. While the subsequent coding process introduces a degree of analytical coherence, variations in source selection and reporting practices remain. Second, press coverage itself is uneven and often incomplete. Many disappearance cases are reported only briefly and without follow-up, limiting the amount of information available for analysis. This results in gaps in the dataset and constrains the ability to reconstruct events in detail. Finally, because the corpus is derived exclusively from press sources, it reflects the selective visibility of cases. Not all disappearances are reported in the media, and those that are reported may not be representative of the broader universe of cases.

4.2 Descriptive Statistics

The main annotated corpus contains information about 575 victims and 2,229 source reports. Figure 2(a) shows the distribution of the reports (documents) associated with each victim. These vary considerably, with a median of 3 and a mean of 4, and a range from a single document to 16. Most victims have between one and three associated reports, while a smaller number are covered by

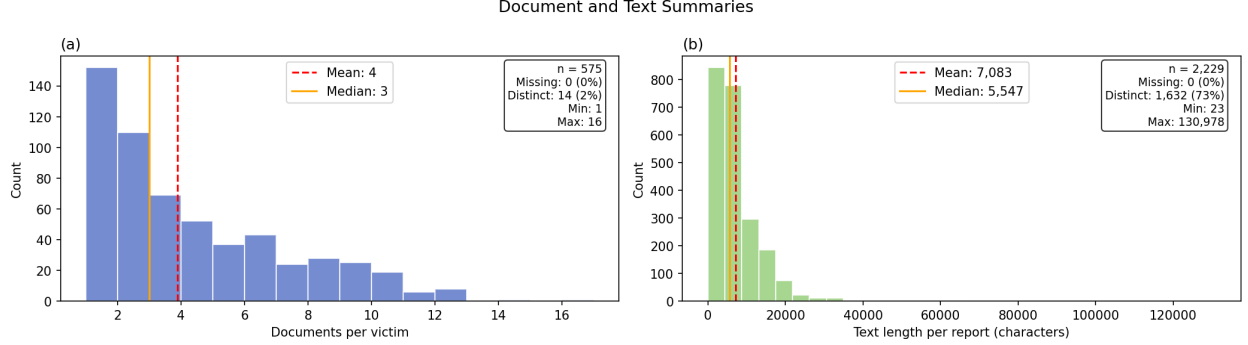


Figure 2: Document and text summaries. (a) Distribution of document counts per victim. (b) Distribution of text length per report.

ten or more. Report length also varies widely across the corpus, with a median of approximately 5,500 characters and a mean of approximately 7,100, but the distribution has a long right tail, with individual reports reaching over 130,000 characters as shown in Figure 2(b). These distributions indicate that both the coverage per victim and the length of individual reports are uneven, which directly encouraged the multi-turn processing pipeline described in the method section. When concatenating the reports for each victim, the total length of the concatenated text will be too long for most LLMs to easily process, which is why we need to use the multi-turn processing pipeline so that corpora can be naturally processed document by document as smaller units.

The codebook itself is extensive, spanning seven sections that cover victim demographics, circumstances of capture, outcome of the disappearance, perpetrator characteristics, judicial proceedings, and observational comments, with several dozen substantive coded variables in total. Not all victims in the corpus have annotations for every coded field. A substantial portion records free text descriptions, dates, or geographic identifiers that cannot be used directly in a classification-based evaluation. Among the categorical fields that remain, many encode details that are specific to a narrow research domain, such as the identity of individual organized crime groups or the type of judicial sentence. The codebook used by the original human coders instructed them to enter 999 when no information was available in the news articles, and as a result many labels carry high rates of missing values across the 575 victims. The degree of missing values varies substantially from one label to another, depending on whether the coder was able to find the information in the news articles for the given victim.

We select a subset of categorical labels that have relatively low rates of missing values and that capture information relevant to a broad audience rather than to a single specialized subfield.

Three labels meet these criteria. The variable / column *Desenlace* records the outcome of the disappearance as reported in the press. Among the 575 victims, 393 have an annotation for this variable, each recording has one of the seven possible categories. The distribution is heavily concentrated. 254 victims are recorded as still disappeared and 91 as found dead, together accounting for nearly 88 percent of annotated cases, while the remaining five categories, including liberation by captors or authorities, found alive, self-liberation, and found without specification of condition, each

contain fewer than 25 cases, as shown in Figure A.4. The *Captura tipo* label records the type of place where the victim was captured, using place categories from the HURIDOCs micro-thesaurus. Among the 575 victims, 243 have an annotation for this field, each recording has one of the eight possible categories. The distribution is heavily concentrated. 90 victims are recorded as means and routes of transport and places of connection and 81 as places related to the victim such as the home or workplace, together accounting for over 70 percent of annotated cases, while the remaining six categories range from 1 to 28 cases, as shown in Figure A.5. The third label that meets these criteria, *Vic grupo social*, records the social group to which the victim belongs. Among the 575 victims, 273 have an annotation for this field, each recording has one of the ten possible categories. The distribution is heavily concentrated. 95 victims are recorded as people who work in service industries, 49 as civil servants, and 45 as students, together accounting for nearly 70 percent of annotated cases, while the remaining seven categories, including professionals, activists, people associated with politics, organized crime, land workers, and others, each contain smaller counts, as shown in Figure A.6.

5 Evaluation

Evaluation of summarization tasks compares the text summary with a gold standard reference text. Given the messy and noisy initial documents, it is difficult to use them as reference texts. The prior work has human annotations classifying the information in the summary text. The evaluation tests whether generated summaries reproduce the same labels as the human annotations, comparing a simple summarization baseline against the proposed method, across major model families, and across model sizes.

5.1 Human Gold Standard

The dataset contains human annotations coded at the victim level, and these annotations are used as ground truth. Should the research use the source documents as reference texts, the output generated from original text will be heavily influenced by the noise, so that the difference between the result of original text and the result from cleaned summaries will be inflated, making the evaluation not reliable. Instead, evaluation proceeds by classifying the generated summary text and comparing the resulting labels against the human annotations, testing whether the summaries preserve the same information that human coders recorded between the results from a direct generative summarization and across multiple models.

The original codebook contains fine-grained categories, and the inherent ambiguity of the source text means that the model and the human coder may generate different but equally defensible results from the same spans. Some distinctions, such as general kidnappings versus special type kidnappings like Levantón, are also difficult for the model without external context plugged in or domain-specific fine-tuning. Evaluating directly against the fine-grained categories would penalize valid model outputs. To reduce the number of categories where multiple valid answers fall into

different classes, the original categories are collapsed into broader groupings before evaluation. The detailed distribution of the collapsed categories is shown in Table 2.

Table 2: Distribution of the three selected labels per victim after merging.

Category	Count
<i>Desenlace</i> (n = 393, missing 182 / 32%, 2 distinct)	
Still disappeared	254
Found	139
<i>Captura tipo</i> (n = 243, missing 332 / 58%, 2 distinct)	
Places related to the victim	81
Public and institutional spaces	162
<i>Vic grupo social</i> (n = 273, missing 302 / 53%, 5 distinct)	
Professionals	145
Civil servants	61
Students	45
Other	19
Organized crime	3

Another issue of the human annotations is the lost documents. Many of the decisions were made when the human coders were able to access more documents than we currently can fetch, either from their sources or from the internet. This means that some of the annotations are not supported by the documents that are available to us. To address this issue, we need to select the annotations that are supported by the documents that are available to us. This is also critical for the goal of this project, which is to relink the annotations to the source text and make them usable for downstream tasks.

5.2 LLM as Evaluators

Recent work has proposed LLM-as-judge evaluators for summarization that compare a generated summary directly against its source text rather than against a human reference summary. SummaC treats factual consistency as a source grounding problem and uses natural language inference over smaller units of the source and the summary to estimate whether the content of the summary is supported by the original document (Laban et al. 2022). G-Eval uses a large language model as a judge and asks it to score generated output through structured prompts and intermediate reasoning steps (Liu et al. 2023). These methods are useful because they move evaluation away from surface similarity and toward the question of whether a summary remains faithful to the source.

In our setting, however, these methods are difficult to use in a meaningful way because the source reports are themselves messy, noisy, and often mixed with irrelevant but topically related material. A source to summary consistency score may therefore reward a system for preserving noise, or penalize a cleaned and focused summary for leaving out distractive content that should not be carried forward. These evaluator does not effectively provide informative evaluation for this

research. A better solution is to adhere to the existing human annotation for evaluation as stated in the previous section.

5.3 Hallucination Evaluation

The hallucination evaluation literature provides an important metric for LLMs in computational social science in a broader sense. The most relevant problem here is input conflicting hallucination, where generated content is not supported by the source text. This is the central risk for our task because the model is asked to extract and summarize information from noisy documents rather than answer open domain questions from background knowledge. However, given that nature of our input text, the hallucination evaluation is not as straightforward as the ones in the literature.

Recent literature generally groups evaluation methods into human evaluation, model based evaluation, and rule based evaluation (Y. Zhang et al. 2025). Human evaluation remains the most reliable because annotators can examine whether each generated claim is supported, unsupported, or irrelevant with respect to the source text (Min et al. 2023). However, given the purpose of our task, the human evaluation does not solve the problem shared with human coder.

LLM based methods then approximate this process in different ways. Large Language Models (LLMs) based methods use an evaluator model to judge factual consistency between the source and the output, sometimes with retrieved evidence (Liu et al. 2023), while rule based methods use structured comparisons such as question answering over the source and the summary, textual entailment style checks, named entity based comparisons, or source grounded reference contrasts (Laban et al. 2022; Lee et al. 2022).

Hallucination evaluation matters in our project because the outcome of the proposed method is extractive summarization that should preserve the information from the source text while leaving out the noise. The generation part opens up the risk of hallucination, so evaluation of hallucination is necessary to ensure the quality of the output (Y. Zhang et al. 2025).

This is not only a language quality issue but also critical for downstream tasks. In political science and computational social science setups, unsupported or even entirely fabricated generated details can change coded variables, weaken traceability, and distort substantive conclusions. A careful evaluation of hallucination therefore contributes to the field by making LLM based extraction more transparent, more reproducible, and more defensible as empirical evidence.

6 Results

6.1 Baseline vs Proposed Method

The baseline for this evaluation is a simple summarization approach that generates a direct summary of each victim’s documents without codebook guidance or structured extraction. The proposed method instead uses the codebook schema to guide extraction and produces label-structured summaries anchored to verbatim evidence spans. Each panel in Figure 3 compares these two approaches on one of the three selected labels, reporting the classification accuracy of the simple

summarization baseline, the extractive summarization method, and the difference between them. Across all three labels and all three models, the proposed method consistently outperforms the baseline, with every difference value being positive. The improvement is largest for outcome, where extractive accuracy reaches 0.80 to 0.84 compared to 0.46 to 0.61 for the baseline, yielding gains between +0.19 and +0.38. Social group shows similarly substantial gains, rising from a baseline range of 0.33 to 0.43 to an extractive range of 0.61 to 0.67. Type of place of capture remains the most difficult label for all models, with extractive accuracy between 0.36 and 0.40, but even here the proposed method improves over the baseline by +0.09 to +0.15. These results indicate that codebook-guided extraction produces summaries that preserve the information needed for downstream classification more reliably than unconstrained summarization across all labels evaluated.

6.2 Model Comparisons

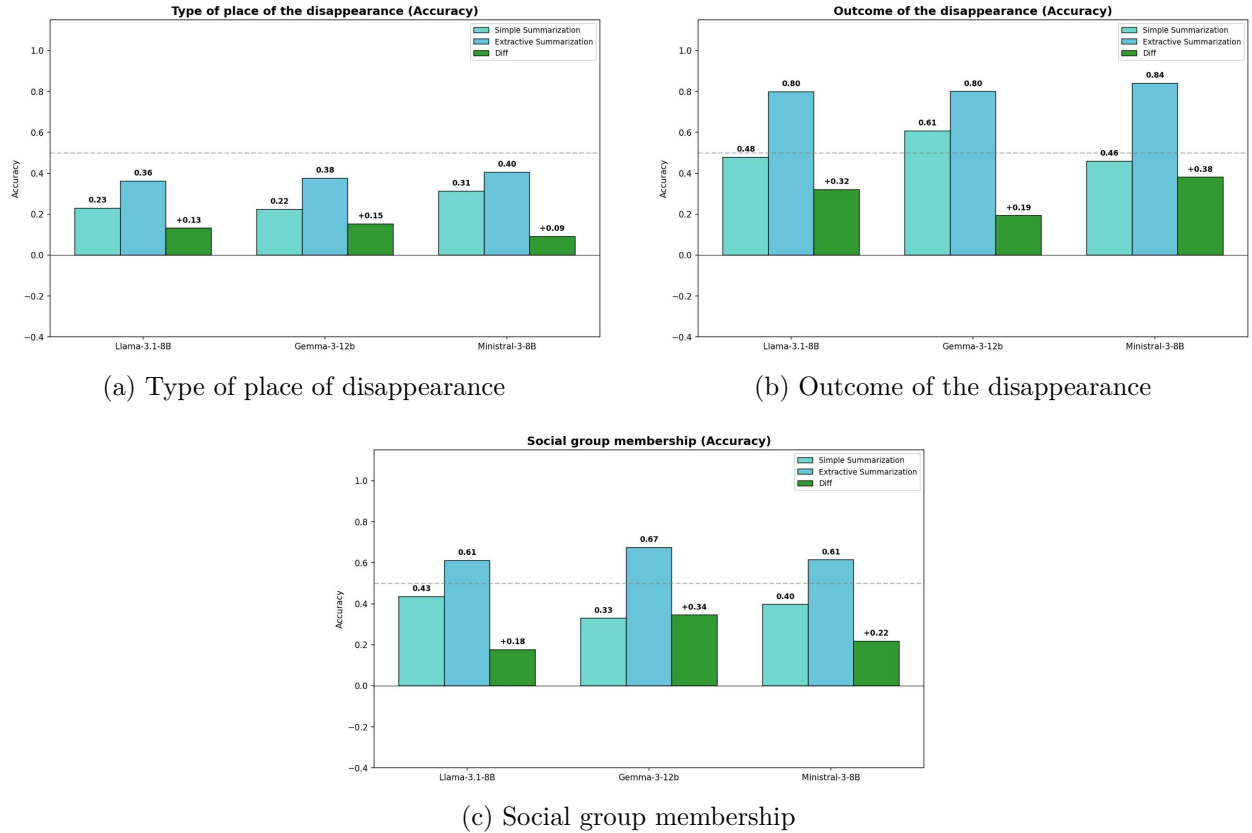


Figure 3: Accuracy by model for three labels in each subplots. Within each subplot, teal bars show the simple summarization baseline, blue bars show extractive summarization, and green bars show the difference between extractive summarization and the baseline. In each subplot, the bars on the left is the results from Llama 3.1 8b, bars on the middle are from Gemma 3 12b, and bars on the right are from Ministral 3 8b

Figure 3 also allows a cross-model comparison, since each panel shows all three models side by

side on the same label. All three models follow the same pattern: the proposed extractive method outperforms the simple summarization baseline on every label, and no model fails to benefit from the structured workflow. The models differ in where they show the largest gains. Gemma 12B achieves its strongest improvement on social group, rising from 0.33 to 0.67 for a gain of +0.34, while Ministral 8B shows its largest gain on outcome, rising from 0.46 to 0.84 for a gain of +0.38. Llama 8B falls between the two, with its largest gain also on outcome at +0.32. For type of place of capture, all three models remain below 0.40 even with extractive summarization, confirming that this is the most difficult label to classify regardless of which model is used. The consistency of the improvement across all three model families suggests that the performance gain originates in the workflow design rather than in the capability of any single model.

The three models also span a range of parameter counts, with Llama 3.1 8B and Ministral 3 8B at 8 billion parameters and Gemma 3 12B at 12 billion. Despite this difference, Gemma does not uniformly outperform the two smaller models. Ministral 8B achieves the highest extractive accuracy on outcome at 0.84 and on type of place of capture at 0.40, while Llama 8B and Ministral 8B both reach 0.61 on social group compared to 0.67 for Gemma. The 4 billion parameter gap does not produce a consistent advantage on any label, which suggests that within the 8B to 12B range the proposed workflow provides comparable benefits regardless of model size.

6.3 Annotation to Documents Mapping

A central question for this pipeline is whether the original human annotations can be traced back to evidence in the available source documents and, where they cannot, whether the constraint is the state of the archive rather than the quality of extraction.

Each figure in this section contains panels that measure the extent to which human annotations can be covered given the source documents that remain available. The cumulative coverage plot tracks the percentage of victim-model pairs that have reached a given support threshold at each turn, where a turn corresponds to processing one additional document from the victim’s record. The coverage of annotation plot shows how many annotated labels each victim carries and whether all source documents for that victim can still be retrieved or whether only a subset remains accessible.

Figure 4 reports these measures for the three selected labels across all three models. Of all victim-model pairs, 16 percent reach full support, meaning every annotated label for that victim is grounded in extracted evidence, while 44 percent reach at least 60 percent support. The coverage of annotation reveals that 381 of the 427 annotated victims retain full document coverage for these selected labels, while the remaining 46 have only partial coverage because some of the documents the original coders consulted are no longer be retrievable. This 11 percent partial-coverage rate is an early indication that the lost documents problem constrains the pipeline even on the best-covered labels, since annotations that were based on missing sources cannot be supported regardless of extraction quality. Among the pairs that do reach their threshold, the large majority arrive at turn zero or turn one, indicating that the first one or two documents already contain most of the recoverable evidence.

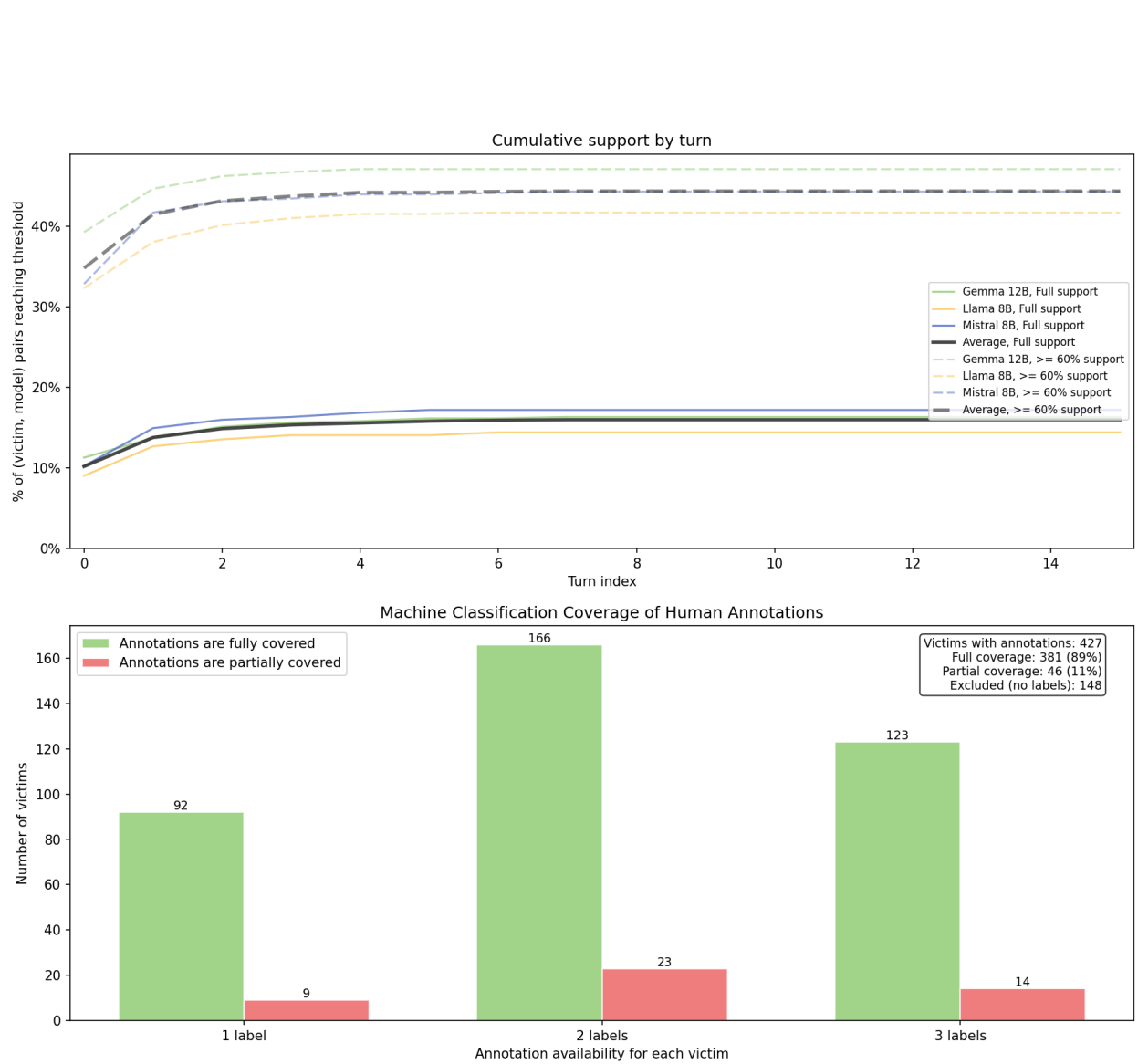


Figure 4: The first plot shows how many turns or documents does a victim needs to achieve the coverage of human annotation; the second plot shows how many victims has their annotations covered, by the availability of the given human annotation.

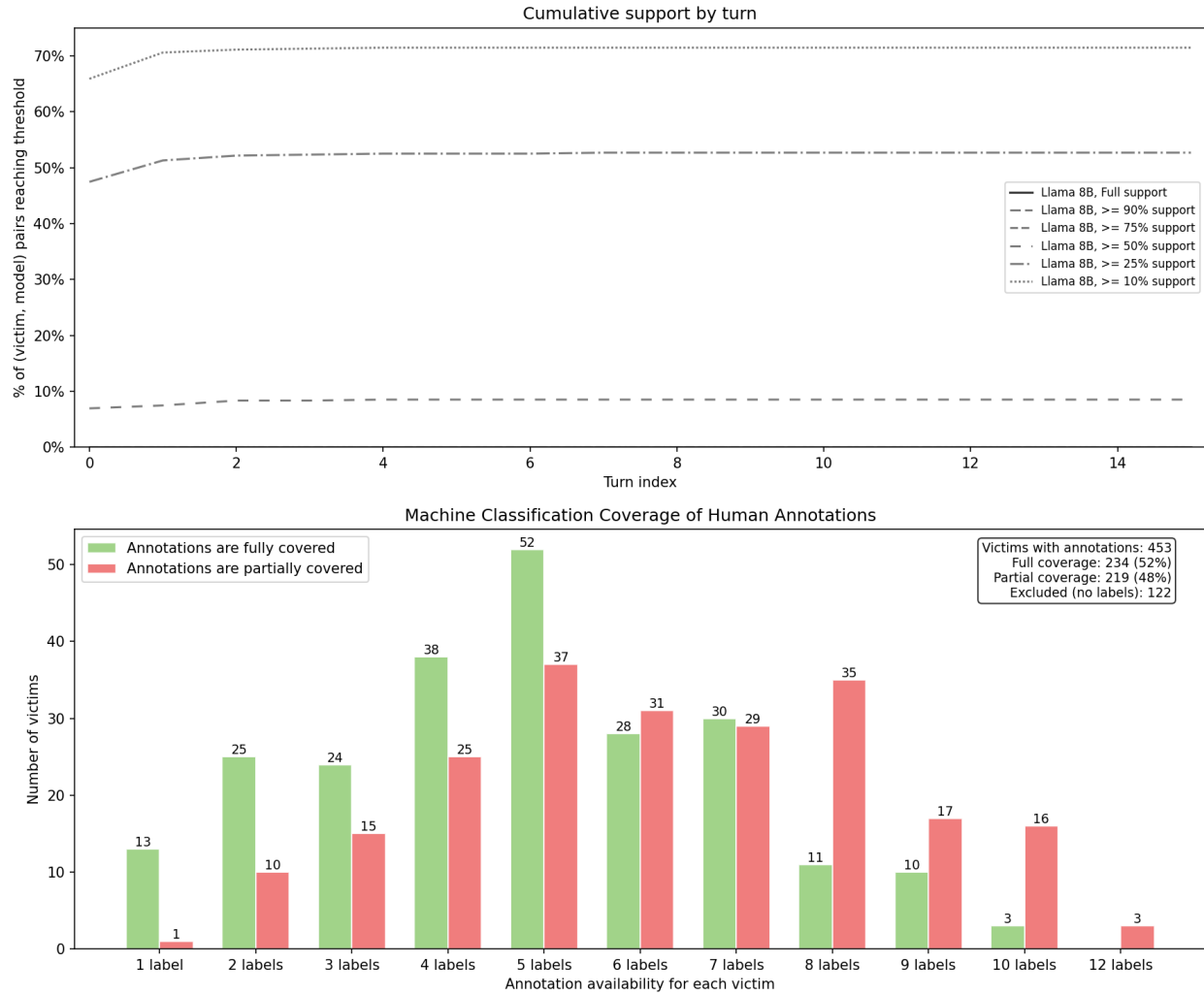


Figure 5: The first plot shows how many turns or documents does a victim needs to achieve the coverage of human annotation; the second plot shows how many victims has their annotations covered, by the availability of the given human annotation.

The three selected labels were chosen for their low missing rates and their broad relevance, but the full codebook contains many more coded fields. Expanding the evaluation to extra categorical labels to include 15 labels in total to tests whether the pipeline can trace annotations that carry higher missing rates and that encode more specialized details. This is a harder test because full support now requires every annotated field for a victim to be grounded in the available text.

Figure 5 reports the same measures for the expanded label set using a single model. Full support drops to approximately 8 percent, which is expected given that every annotated label must now be matched simultaneously. At lower thresholds the pipeline still recovers evidence for a substantial share of annotations: 71 percent of pairs reach at least 10 percent support and 53 percent reach at least 25 percent. The critical constraint is visible in the machine classification coverage panel, where only 234 of the 453 annotated victims retain full document coverage, falling from 89 percent in the three-label setting to 52 percent when all labels are considered. The remaining 219 victims have only partial coverage because news articles that were available to the original human coders are no longer be retrievable from their sources or from the internet. This is the direct quantification of the lost documents problem introduced in the evaluation design that the pipeline can only process the documents that remain in the accessible archive, and any annotation that depended on a source that has since disappeared sets a hard ceiling on the support that can be recovered. The turn-by-turn recording that the pipeline produces makes this ceiling visible and locatable. For the annotations that can be traced, most pairs reach their threshold within the first two documents. For those that cannot be fully traced, the recording identifies precisely where the evidence trail ends, producing a concrete inventory of what the pipeline recovers and what remains beyond reach due to the state of the archive. This inventory is the practical contribution of the tracing dimension: rather than treating missing support as an unquantified limitation, the pipeline distinguishes recoverable annotations from structurally unrecoverable ones and documents the boundary between them.

victim_id	label_key	span	offset
Veracruz_J J R R	desenlace	“Hasta la fecha, sigue en la búsqueda, se ha convertido en detective. . .”	2480
Veracruz_J J R R	vic_grupo_social	“Carmelo estudiaba arquitectura en una escuela privada de diseño. Su círculo de amigos era muy grande. . .”	2691
N.L._Fabián H V	desenlace	“En el expediente no hay registros del plagio ni testimonios.”	2585
Veracruz_María Victoria C T	captura_tipo	“Según Lorena, su madre y hermano fueron sustraídos de un taxi entre la calle 13 y avenida 11, frente al DIF Municipal, el pasado 17 de abril.”	1254
N.L._Fabián H V	captura_tipo	“Los perseguidores sustrajeron al empresario del vehículo siniestrado y se lo llevaron con rumbo desconocido en otra unidad, un Audi.”	4109

Table 3: Sample extracted spans per codebook field. Each span is a verbatim excerpt from the source document; the offset records the character position where the span begins in the original text.

Tables 3 and 4 show abbreviated examples from the full pipeline output for two of its recorded artifacts. Table 3 presents a sample of extracted spans. Each row records the victim identifier, the

turn	doc_id	info_found	labels found	summary (excerpt)
0	_1	FALSE	—	(no summary produced)
1	_2	TRUE	desenlace, vic_grupo_social	“Cuatro personas originarias de Sinaloa desaparecieron... fueron secuestrados por policías preventivos... y entregados a un cártel...”
2	_6	FALSE	—	(previous summary preserved unchanged)
3	_8	TRUE	desenlace, vic_grupo_social, captura_tipo	“...siete jóvenes detenidos por policías locales y entregados a Los Zetas... la policía local colaboraba con el crimen organizado...”

Table 4: Summary evolution across four turns for Coahuila_Ezequiel C T. Turns 0 and 2 process noisy documents that yield no information; the pipeline preserves the previous summary unchanged. Turns 1 and 3 extract new evidence and update the running summary.

codebook field that the span addresses, the verbatim passage extracted from the source document, and the character offset where that passage begins in the original text. Table 4 presents the turn-by-turn processing trace for a single victim across four documents. Each row records the turn index, the document identifier, whether relevant information was found in that document, which codebook labels were identified, and an excerpt of the running summary at that point. Both tables are abbreviated selections from a larger output that the pipeline produces for every victim in the corpus.

In Table 4, the pipeline processes four documents sequentially for one victim. At turns zero and two, the documents contain no relevant information for the victim. These are the noisy, irrelevant, or damaged pages that are common in web-scraped corpora, such as error pages, log in pages, or unrelated articles scraped alongside the target content. When the pipeline encounters such documents, it preserves the existing summary unchanged rather than letting irrelevant content dilute or degrade the evidence already collected. At turns one and three, the pipeline finds relevant information and updates the running summary accordingly. The set of supported codebook labels grows from two at turn one, where the pipeline identifies outcome and social group, to three at turn three, where it additionally identifies type of place of capture. This progressive enrichment shows that evidence for different codebook fields can be distributed across multiple documents and that the stateful design accumulates labels as informative documents are encountered. In Table 3, the character offset for each extracted span records the exact position in the source document where the evidence begins. A researcher can return to the original text at the recorded offset and verify that the extracted passage appears at that location, making each piece of evidence independently checkable against the source.

The combination of verbatim spans, character offsets, and turn-by-turn processing traces means the pipeline output is auditable at the level of individual evidence claims. A researcher can follow the chain from a codebook label to the extracted passage that supports it to its character position in the source document to the specific processing turn that produced it. At the aggregate level, the support coverage analysis identifies which annotations fall within reach of the available archive and which

are structurally unrecoverable because of the lost documents problem. At the instance level, the extracted spans and turn traces document precisely what was recovered and where the evidence trail ends. These two dimensions produce a complete account of the annotation-to-document mapping: the pipeline does not only classify but records the evidence path for each annotation it can trace and flags the structural reason for each one it cannot.

7 Discussion

7.1 Compute Time vs Human Coding

An important implication of the proposed workflow is that it changes the cost structure of working with inherited annotations. Recreating the earlier coding process would require retraining coders, rereading multiple reports for each victim, and reconstructing decisions that were never formally preserved, whereas the pipeline can process the same material in hours while recording each turn of evidence extraction and summary updating. The gain is therefore not only speed, but a different use of scarce expert labor. Instead of spending weeks on repetitive document cleaning, reconciliation, and aggregation, researchers can direct human effort toward validating difficult cases, resolving ambiguous categories, and interpreting substantive patterns in the data, while the repetitive parts of the task are taken care of by the LLMs. For projects that inherit valuable but imperfectly organized annotations, this makes reuse of prior coding work far more feasible than full manual reconstruction.

7.2 Scalability to Other Messy Corpora

A broader value of the proposed method lies in the type of problem it addresses rather than in the specific corpus used here. Many research projects inherit messy text and annotation collections in which reports are organized for one purpose, annotations were created for another, and the link between coded variables and supporting evidence was never formally retained. In such settings, the transferable contribution is the workflow design itself the codebook-guided extraction, evidence-linked summaries, and stateful updating across multiple documents. This portability is possible because the workflow is designed to be configurable rather than hardcoded. Separate configuration files and taxonomies can define the local labels, extraction targets, and field structure for a new corpus while leaving the underlying pipeline logic unchanged. Adapting the method to a new corpus would therefore require changing the task schema and prompts to fit the local annotation system, but not redesigning the full logic of the pipeline. This suggests that the approach can generalize to other legacy corpora where researchers need to recover usable and traceable structured text from noisy materials without repeating the original annotation process from the beginning.

7.3 Fine-Tuning on Pipeline Output

The fine-tuning dataset is constructed as follows. For each victim, the pipeline produces a final structured summary in which every codebook field is populated with verbatim spans extracted from

the source reports. These summaries are paired with the corresponding human annotations for fields such as case outcome (*desenlace*), victim social group (*vic_grupo_social*), and type of place of capture (*captura_tipo*). Each training instance therefore consists of a clean, evidence-grounded summary as input and one or more categorical labels as the target. Because the summaries are already organized label by label, the fine-tuning input can be formatted either as a full summary for multi-label prediction or as a per-field segment for single-label classification, depending on the downstream task design. In the current corpus, the annotated subset provides up to 393 labeled instances for *desenlace*, 273 for *vic_grupo_social*, and 243 for *captura_tipo*, each with the corresponding pipeline-cleaned summary as the text input.

To validate this empirically, we compare classification performance when models are trained on the original raw messy text versus the pipeline-cleaned summaries, using stratified 5-fold cross-validation across all three selected labels. Traditional TF-IDF bag-of-words classifiers favor the raw text due to its greater lexical volume: the mean raw document length per victim is approximately 28,000 characters compared to roughly 700 characters for the summaries, providing more discriminative n-grams even though many come from navigation menus or unrelated content. However, this advantage disappears when the classifier operates semantically. Table 5 reports results using multilingual sentence-transformer embeddings (paraphrase-multilingual-MiniLM-L12-v2) followed by logistic regression. The pattern reverses on two of the three labels: for *desenlace*, summaries achieve a macro F1 of 0.723 compared to 0.656 on raw text, a gain of +0.067; for *vic_grupo_social*, summaries reach 0.484 compared to 0.418, a gain of +0.066. For *captura_tipo*, the two inputs perform comparably at 0.587 and 0.600 respectively, consistent with this being the most difficult label across all experiments. The κ values follow the same pattern, with summaries showing stronger agreement on *desenlace* (0.450 vs. 0.315) and *vic_grupo_social* (0.376 vs. 0.335).

Table 5: Classification performance (stratified 5-fold CV) using multilingual sentence-transformer embeddings with logistic regression on raw messy text vs. pipeline-cleaned summaries. Best macro F1 per label shown in bold. N denotes the number of victims with both a valid label and summary.

Label (N)	Input	Accuracy	Macro F1	κ
Outcome (381)	Raw text	0.701 $\pm$ 0.041	0.656 $\pm$ 0.048	0.315 $\pm$ 0.095
	Summary	0.758 $\pm$ 0.023	0.723 $\pm$ 0.029	0.450 $\pm$ 0.055
Social Group (263)	Raw text	0.620 $\pm$ 0.042	0.418 $\pm$ 0.047	0.335 $\pm$ 0.078
	Summary	0.639 $\pm$ 0.037	0.484 $\pm$ 0.065	0.376 $\pm$ 0.069
Place of Capture (241)	Raw text	0.697 $\pm$ 0.061	0.600 $\pm$ 0.075	0.232 $\pm$ 0.136
	Summary	0.697 $\pm$ 0.048	0.587 $\pm$ 0.091	0.211 $\pm$ 0.157

These results confirm that when models process text semantically rather than lexically, the clean and concentrated pipeline summaries outperform the noisy raw documents. Since LLM fine-tuning operates on the semantic end of this spectrum, Table 5 provides the more relevant baseline for the downstream use case. Because the pipeline records which evidence spans support each label, training instances can be augmented with span-level supervision, enabling fine-tuned models to

learn not only the correct label but also which parts of the text are relevant to that label, a form of rationale-augmented training.

The fine-tuning procedure itself follows standard practice for instruction-tuned language models. The pipeline summaries and their associated labels are formatted as prompt–completion pairs, where the prompt contains the structured summary text and the codebook field definition, and the completion contains the target label. Parameter-efficient methods such as LoRA can be applied to adapt smaller open-source models, for instance a 7B or 8B parameter model, on this labeled data without requiring full-weight training. Because the summaries are already constrained to the codebook schema and written in the same structured format regardless of which extraction model produced them, the fine-tuning data is consistent in format and length, which simplifies training and reduces variance across batches.

The practical implication is that the pipeline enables a two-stage division of labor in which large, general-purpose models handle the expensive extraction and summarization at inference time, and the resulting clean outputs are used to train smaller, cheaper models that can be deployed for classification at scale without requiring continued access to a large model endpoint. Once fine-tuned, these smaller models can classify new summaries produced by the same pipeline for new victims or new corpora at a fraction of the inference cost. This fine-tuning pathway also opens the possibility of iterative improvement: as domain experts review and correct the pipeline classifications, those corrections can be folded back into the training data to produce progressively better task-specific models. The pipeline therefore functions not only as a one-time cleaning tool but as infrastructure for generating and refining supervised training data from legacy corpora that were never originally annotated at the granularity that modern fine-tuning workflows require.

7.4 Downstream Tasks

When developing new research projects using the UMN dataset, we are particularly interested in analyzing how violence is portrayed, described, or obscured through language in public reporting, especially in ways that prevent the systematic identification of human rights and international crimes. Our work focuses on identifying the use of euphemisms, ambiguous terminology, and narrative framing that mask patterns of violence and make it more difficult to establish the widespread or systematic nature of abuses—an essential requirement for proving international crimes before courts. By analyzing large corpora of media reports and related documentation, we aim to understand how linguistic practices can shape the visibility of violence and influence whether certain acts are recognized as human rights violations.

One specific focus of our research using the UMN dataset concerns the terminology that emerged during Mexico’s so-called “War on Drugs,” declared by President Felipe Calderón in 2006. During this period, journalists increasingly used the term “levantón” (“big lift”) to describe events involving the deprivation of liberty followed by the disappearance of individuals whose fate and whereabouts remained unknown. Although some journalists now employ more accurate terminology, many reports still frame disappearances primarily as ordinary criminal acts rather than as serious human rights

violations, often overlooking the possible role of the state through action, tolerance, or omission. Our research seeks to examine how such language contributes to the normalization or depoliticization of enforced disappearance in public discourse.

Presenting disappearances as isolated crimes creates several analytical and legal problems. First, it reduces the event to a single act of deprivation of liberty rather than recognizing enforced disappearance as a continuous and ongoing crime that persists as long as the person’s fate remains unknown. Second, it obscures the multiple human rights violations involved, including violations of the rights to life, personal integrity, and juridical personality, and ignores the broader circle of victims affected by the disappearance. Finally, it conceals the possible involvement of the state, which is central to the legal definition of enforced disappearance. The use of terms like “levantón” is particularly problematic because it originates in the lexicon of organized crime and functions as a euphemism that masks what may in fact be enforced disappearance. Through systematic analysis of this language, our research seeks to reveal how linguistic framing can obscure patterns of violence that are essential for legal accountability and human rights documentation.

8 Conclusion and Future Work

The central obstacle in working with inherited messy text is not noise alone, but the broken connection between prior human coding and the source evidence on which that coding depended. The proposed codebook-guided extractive summarization workflow addresses that gap by producing clean, evidence-linked summaries that remain traceable to the original reports and usable for downstream language model tasks. It offers a practical way to preserve and reactivate costly human judgment that would otherwise remain trapped inside legacy annotations, while shifting scarce expert labor away from repetitive reconstruction and toward validation, interpretation, and substantive analysis.

Future work can build on this recovered representation as research infrastructure for organizing text datasets by a schema driven by codebook definitions for cumulative human rights research, future investigations, historical reconstruction, and accountability over time. In that role, it can help sustain shared evidence across long term collaborative projects and make structured documentation more usable for local organizations, independent researchers, victims, and affected communities, and could be applied to existing efforts such as the Notre Dame Legacy Project’s archive of the Colombian Truth Commission (University of Notre Dame [2026](#)).

References

- Biester, Fabian, Mohamed Abdelaal, and Daniel Del Gaudio (2024). “LLMClean: Context-Aware Tabular Data Cleaning via LLM-Generated OFDs”. In: *Advances in Databases and Information Systems (ADBIS)*. Vol. 2186, pp. 68–78. DOI: [10.48550/arXiv.2404.18681](https://doi.org/10.48550/arXiv.2404.18681).
- Bohannon, Philip et al. (2007). “Conditional Functional Dependencies for Data Cleaning”. In: *Proceedings of the 23rd International Conference on Data Engineering (ICDE)*, pp. 746–755.
- Brandt, Patrick T. et al. (2025). “Extractive versus Generative Language Models for Political Conflict Text Classification”. In: *Political Analysis*, pp. 1–29. DOI: [10.1017/pan.2025.10027](https://doi.org/10.1017/pan.2025.10027).
- Choi, Juhwan et al. (2024). “Multi-News+: Cost-efficient Dataset Cleansing via LLM-based Data Annotation”. In: *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. DOI: [10.48550/arXiv.2404.09682](https://doi.org/10.48550/arXiv.2404.09682).
- Dagdelen, John et al. (2024). “Structured Information Extraction from Scientific Text with Large Language Models”. In: *Nature Communications* 15, p. 1418. DOI: [10.1038/s41467-024-45563-x](https://doi.org/10.1038/s41467-024-45563-x).
- Ding, Zhicheng et al. (2024). *Data Imputation using Large Language Model to Accelerate Recommendation System*. arXiv:2407.10078. DOI: [10.48550/arXiv.2407.10078](https://doi.org/10.48550/arXiv.2407.10078).
- Dubey, Abhimanyu et al. (2024). *The Llama 3 Herd of Models*. arXiv:2407.21783. DOI: [10.48550/arXiv.2407.21783](https://doi.org/10.48550/arXiv.2407.21783).
- Escott, Paul D. (2018). *Quantitative Data Coded from the Federal Writers’ Project Slave Narratives, United States, 1936–1938*. Inter-university Consortium for Political and Social Research [distributor]. Dataset. DOI: [10.3886/ICPSR36381.v1](https://doi.org/10.3886/ICPSR36381.v1). URL: <https://doi.org/10.3886/ICPSR36381.v1>.
- Fan, M. et al. (2024). “Cost-Effective In-Context Learning for Entity Resolution: A Design Space Exploration”. In: *Proceedings of the 40th IEEE International Conference on Data Engineering (ICDE)*. DOI: [10.1109/ICDE60146.2024.00284](https://doi.org/10.1109/ICDE60146.2024.00284).
- Frantar, Elias et al. (2024). *MARLIN: Mixed-Precision Auto-Regressive Parallel Inference on Large Language Models*. arXiv:2408.11743.
- Gemma Team (2025). *Gemma 3 Technical Report*. arXiv:2503.19786. DOI: [10.48550/arXiv.2503.19786](https://doi.org/10.48550/arXiv.2503.19786).
- Gibbs, Graham R., Dawn Clarke, and Andrew Teal (2010). *The Victoria Climbié Corpus*. University of Huddersfield Repository. Dataset. URL: <https://eprints.hud.ac.uk/id/eprint/8886>.
- He, Xinrui et al. (2025). “LLM-Forest: Ensemble Learning of LLMs with Graph-Augmented Prompts for Data Imputation”. In: *Findings of the Association for Computational Linguistics (ACL)*. DOI: [10.18653/v1/2025.findings-acl.361](https://doi.org/10.18653/v1/2025.findings-acl.361).
- Hurtado Bodell, Miriam, Måns Magnusson, and Sophie Mützel (2022). “From Documents to Data: A Framework for Total Corpus Quality”. In: *Socius: Sociological Research for a Dynamic World* 8, p. 23780231221135523. DOI: [10.1177/23780231221135523](https://doi.org/10.1177/23780231221135523).

- Kwon, Woosuk et al. (2023). “Efficient Memory Management for Large Language Model Serving with PagedAttention”. In: *Proceedings of the 29th Symposium on Operating Systems Principles (SOSP)*. DOI: [10.1145/3600006.3613165](https://doi.org/10.1145/3600006.3613165).
- Laban, Philippe et al. (2022). “SummaC: Re-Visiting NLI-based Models for Inconsistency Detection in Summarization”. In: *Transactions of the Association for Computational Linguistics* 10, pp. 163–177. DOI: [10.1162/tacl_a_00453](https://doi.org/10.1162/tacl_a_00453).
- Lee, Nayeon et al. (2022). *Factuality Enhanced Language Models for Open-Ended Text Generation*. arXiv:2206.04624. DOI: [10.48550/arXiv.2206.04624](https://doi.org/10.48550/arXiv.2206.04624).
- Li, Aodong et al. (2024). “Anomaly Detection of Tabular Data Using LLMs”. In: *CoRR* abs/2406.16308. DOI: [10.48550/arXiv.2406.16308](https://doi.org/10.48550/arXiv.2406.16308).
- Liu, Yang et al. (2023). “G-Eval: NLG Evaluation using GPT-4 with Better Human Alignment”. In: *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pp. 2511–2522. DOI: [10.18653/v1/2023.emnlp-main.153](https://doi.org/10.18653/v1/2023.emnlp-main.153).
- Ma, Chuangtao et al. (2025). “Knowledge Graph-based Retrieval-Augmented Generation for Schema Matching”. In: *CoRR* abs/2501.08686. DOI: [10.48550/arXiv.2501.08686](https://doi.org/10.48550/arXiv.2501.08686). URL: <https://arxiv.org/abs/2501.08686>.
- Ma, Luyi et al. (2023). “LLMs with User-Defined Prompts as Generic Data Operators for Reliable Data Processing”. In: *Proceedings of the IEEE International Conference on Big Data (BigData)*. DOI: [10.1109/BigData59044.2023.10386472](https://doi.org/10.1109/BigData59044.2023.10386472).
- Min, Sewon et al. (2023). “FActScore: Fine-grained Atomic Evaluation of Factual Precision in Long Form Text Generation”. In: *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pp. 12076–12100. DOI: [10.18653/v1/2023.emnlp-main.741](https://doi.org/10.18653/v1/2023.emnlp-main.741).
- Mistral AI (2026). *Minstral 3*. arXiv:2601.08584. DOI: [10.48550/arXiv.2601.08584](https://doi.org/10.48550/arXiv.2601.08584).
- Molina, Ignacio, José Morales, and Brian Keith (2025). “Web Scraping Chilean News Media: A Dataset for Analyzing Social Unrest Coverage (2019–2023)”. In: *Data* 10.11, p. 174. DOI: [10.3390/data10110174](https://doi.org/10.3390/data10110174).
- Ni, Wei et al. (2024). “IterClean: An Iterative Data Cleaning Framework with Large Language Models”. In: *Proceedings of the ACM Turing Award Celebration Conference – China (ACM TUR-C)*.
- Olsen, Helene et al. (Mar. 2024). “Socio-political Events of Conflict and Unrest: A Survey of Available Datasets”. In: *Proceedings of the 7th Workshop on Challenges and Applications of Automated Extraction of Socio-political Events from Text (CASE 2024)*. Ed. by Ali Hürriyetoglu et al. St. Julians, Malta: Association for Computational Linguistics, pp. 40–53. DOI: [10.18653/v1/2024.case-1.5](https://doi.org/10.18653/v1/2024.case-1.5). URL: <https://aclanthology.org/2024.case-1.5/>.
- OpenRefine Community (2025). *OpenRefine: A Power Tool for Working with Messy Data*. <https://openrefine.org>. Accessed: 2025-11-05.
- Peeters, Ralph, Aaron Steiner, and Christian Bizer (2025). *Entity Matching using Large Language Models*. arXiv:2310.11244. DOI: [10.48550/arXiv.2310.11244](https://doi.org/10.48550/arXiv.2310.11244).

- Rahm, Erhard (2011). “Towards Large-Scale Schema and Ontology Matching”. In: *Schema Matching and Mapping*. Springer, pp. 3–27.
- Sokol, Anna et al. (2025). “Ventana a la Verdad (Window to the Truth): A Chatbot Application for Navigating The Colombian Truth Commission’s Archives”. In: *Proceedings of the Eighteenth ACM International Conference on Web Search and Data Mining (WSDM)*. ACM. DOI: [10.1145/3701551.3704123](https://doi.org/10.1145/3701551.3704123).
- University of Notre Dame (2026). *The Legacy Project: Preserving and Engaging the Digital Archive of the Colombian Truth Commission*. <https://legacyproject.nd.edu/>. Accessed: 2026-05-06.
- Wang, T. et al. (2025). *Match, Compare, or Select? An Investigation of Large Language Models for Entity Matching*. arXiv:2405.16884. DOI: [10.48550/arXiv.2405.16884](https://doi.org/10.48550/arXiv.2405.16884).
- Yan, Mengyi et al. (2024). “GIDCL: A Graph-Enhanced Interpretable Data Cleaning Framework with Large Language Models”. In: *Proceedings of the ACM on Management of Data* 2.6, 236:1–236:29. DOI: [10.1145/3698811](https://doi.org/10.1145/3698811).
- Zhang, Haochen et al. (2024). “Large Language Models as Data Preprocessors”. In: *VLDB Workshops*. DOI: [10.48550/arXiv.2308.16361](https://doi.org/10.48550/arXiv.2308.16361).
- Zhang, Yue et al. (2025). *Siren’s Song in the AI Ocean: A Survey on Hallucination in Large Language Models*. arXiv:2309.01219. DOI: [10.48550/arXiv.2309.01219](https://doi.org/10.48550/arXiv.2309.01219).
- Zhou, Wei et al. (2026). *Can LLMs Clean Up Your Mess? A Survey of Application-Ready Data Preparation with LLMs*. arXiv:2601.17058. DOI: [10.48550/arXiv.2601.17058](https://doi.org/10.48550/arXiv.2601.17058).

A Appendix

A.1 Ethical Considerations

Masking victim names in the dataset is an important measure to protect the privacy, dignity, and security of victims and their families. Cases involving enforced disappearance and other forms of violence often remain politically sensitive and may involve powerful actors, including criminal organizations or state agents. Publicly circulating identifiable personal information can expose victims’ relatives to intimidation, retaliation, or renewed stigmatization. By anonymizing names while preserving the relevant contextual data—such as dates, locations, and types of violations—the dataset allows researchers to analyze patterns of violence without placing individuals or families at additional risk.

Anonymization also reflects broader ethical standards in human rights research and documentation, which prioritize the protection of victims over the unrestricted dissemination of personal information. Masking identities helps prevent the re-traumatization of families and respects their right to control how their loved ones’ stories are used and shared. The dataset remains analytically valuable because the key variables needed to study trends, language, and patterns of violence are preserved, enabling rigorous research while maintaining responsible safeguards for those directly affected by the crimes being documented.

Página no encontrada - Agencia de Noticias IRZA Saltar al contenido Agencia de Noticias IRZA desde 1992 Menú INICIO GUERRERO CHILPANCINGO ACAPULCO COSTA CHICA COSTA GRANDE CENTRO MONTAÑA NORTE TIERRA CALIENTE SIERRA CONGRESO GOBIERNO DEL ESTADO POLÍTICA POLICIACA DEPORTES NOTA ROJA x Buscar: Botón de búsqueda ¡Vaya! No se puede encontrar esa página. No hemos encontrado nada aquí. ¿Qué tal si pruebas el buscador? 404 error Buscar: Botón de búsqueda Gobierno del estado GOBIERNO DEL ESTADO Evelyn Salgado cumple con la salud de la Montaña y la Sierra entregan ambulancias a Tlapa y Leonardo Bravo 12 de septiembre de 2025 0 Facebook Twitter Email Compartir Chilpancingo, Gro.- En un acto de colaboración y compromiso social con la salud de las familias de Guerrero, el subsecretario de Desarrollo Político y Leer más Refuerza el gobierno de Guerrero comunicación y cooperación con autoridades fronterizas de los Estados Unidos 12 de septiembre de 2025 0 Evalúa Evelyn Salgado condiciones meteorológicas, fortalece cultura de la prevención en Guerrero y alista participación en Simulacro Nacional 12 de septiembre de 2025 0 El Bulevar de Las Naciones se ilumina para celebrar las fiestas patrias 12 de septiembre de 2025 0 Guerrero, el destino para disfrutar este fin de semana patrio 12 de septiembre de 2025 0 Se aleja Depresión Tropical Trece-E de Guerrero; Acapulco tendrá buen clima desde esta noche para celebrar las fiestas patrias 12 de septiembre de 2025 0 Guerrero pone orden en permisos y licencias; se blindo con tecnología contra el robo de autos: Evelyn Salgado 10 de septiembre de 2025 0 Reducirán el uso de plásticos de un solo uso en oficinas del sector educativo: SEG 10 de septiembre de 2025 0 Congreso CONGRESO POLÍTICA No hay distanciamiento entre Morena y el PT en el Congreso local: Urióstegui 10 de septiembre de 2025 0 Facebook Twitter Email Compartir Fernando Polanco Ochoa/IRZACHilpancingo, Gro.- El coordinador del Grupo Parlamentario de Morena del Congreso local, Jesús Urióstegui García, rechazó un

Figure A.1: Example of an HTTP error page in the scraped corpus.

5/10/2019

Liberan a Matatistas para ejecutarle orden de aprehensión | INFO7

 Info 7

- [SERVICIOS](#)  INFO 7
 - [NOTICIAS](#)
 - [REPORTE CIUDADANO](#)
 - [FONDOS DE PANTALLA](#)
 - [CONTACTO](#)
- [SECCIONES](#)  INFO 7
 - [VIDEOS](#)
 - [LOCALES](#)
 - [NACIONAL](#)
 - [INTERNACIONAL](#)
 - [ECONOMÍA](#)
 - [CLIMA](#)
 - [DEPORTES](#)
 - [ESPECTÁCULOS](#)
 - [VIDA Y ESTILO](#)
 - [EDITORIAL](#)

Figure A.2: Example of ancillary website formatting interleaved with report content.

Entradas relacionadas



Nuevo ataque a balazos contra policías: hay tres heridos



“Desapareció” joven empresario de Piedras Negras; piden ayuda para encontrarlo



“Piratea” y se adjudica Javier Guerrero fotos de eventos de Miguel Riquelme

← Concentran en Saltillo a 4 elementos de Fuerza Coahuila por trata de migrantes

Jóvenes son presente y futuro, pero hay que saberlos apuntalar: Martha Garay →

Figure A.3: Example of distractive but topically adjacent content appended to a report.



Figure A.4: Distribution of desenlace categories per victim before merging.

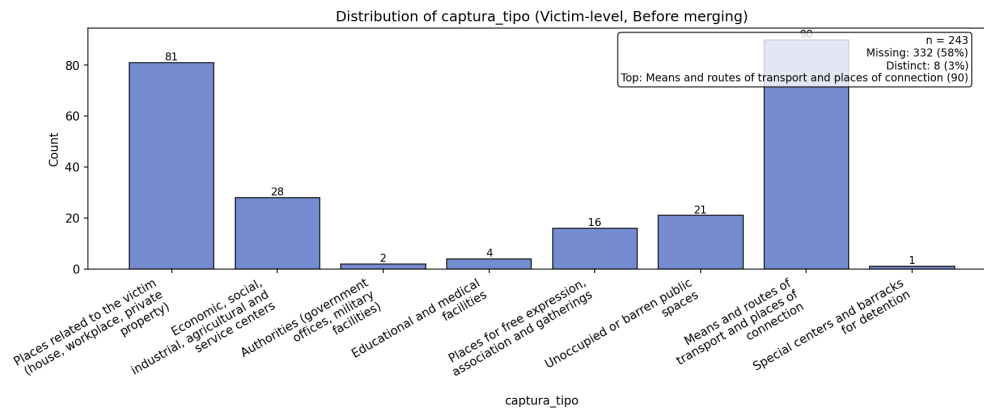


Figure A.5: Distribution of captura tipo categories per victim before merging.

A.2 Rule-based Preprocessing and its Limits

To address these noise sources, the authors attempted to rescrape the original URLs to obtain cleaner versions of the documents. However, a portion of the original URLs were no longer accessible at the time of rescraping. For those that remained available, the structural issues persisted, as the noise originates from the design of the source websites themselves rather than from the scraping process. Navigation menus, comment sections, and related article modules are integral parts of the page layout and are captured by any general-purpose scraper that retrieves the full page body.

The authors also attempted rule-based preprocessing to remove known sources of noise, targeting HTML tags, residual markup, and other structural elements. While this reduced formatting noise, it could not reliably eliminate all types of errors such as navigation text or other irrelevant information, because each webpage were designed differently. As for related content such as suggested articles or comment threads which appear as valid plain text, are almost indistinguishable from article content by pattern-matching rules. Processing them requires substantive human hours and is not scalable. Approaches such as optical character recognition (OCR), which convert image-based documents to machine-readable text, are not applicable here, as the documents were already scraped as plain text. The noise in this corpus is not an accessibility problem but a content boundary or scope problem, and no rule-based or format-conversion method can reliably locate where the article body ends and extraneous content begins without semantic understanding of the text.

A.3 Limitations

One limitation of this study is how performance is evaluated against a codebook where categories are not always sharply separable in the source text. Some reports contain descriptions that can reasonably support more than one label, so disagreement between the model output and the human annotation does not always mean that the summary failed to preserve the relevant information. In these cases, lower accuracy can partly reflect ambiguity in the source material and overlap in the taxonomy, especially for labels where valid interpretations remain close to one another. We mitigate part of this problem by merging some label options, but that adjustment also reduces category detail and makes the comparison less a direct reproduction of the original coding scheme. Even though the proposed workflow is not a full reconstruction of the original coding scheme, it is still a practical way to reconnect prior annotations to the source texts and recover traceable evidence for downstream use.

A.4 Error analysis

Three identifiable sources of error affect the pipeline at different stages. The first is document-level noise from error pages, navigation menus, advertising banners, and topically adjacent but irrelevant content that pervade the web-scraped corpus. LLM-based data standardization and error processing can handle such noise through contextual extraction and generation rather than rule-based methods. The pipeline applies this principle by using the codebook schema to focus extraction on relevant

fields, so that noisy content is filtered during the extraction step itself rather than requiring a separate preprocessing stage.

The second source is missing evidence from the lost documents problem. When source articles consulted by the original coders are no longer retrievable, annotations that depended on those sources cannot be grounded regardless of extraction quality. The pipeline addresses this by recording which annotations remain unsupported after all available documents have been processed and by distinguishing structural absence of the source from the archive from extraction failure through the support ceiling and turn-by-turn recording. This makes the gap attributable and quantifiable rather than hidden within the overall accuracy figures.

The third and likely largest source of error is ambiguity in the categorization of certain texts. Some labels have options that overlap in meaning or that require contextual judgments the source text does not resolve unambiguously. When a report can reasonably support more than one category, the human coder and the model may arrive at different but defensible assignments, and this disagreement registers as a classification error even though the summary correctly preserved the relevant information. These failure modes being knowable and separable itself is a contribution to methodological transparency in computational social science, because it lets researchers make informed choices about when and how to use the output.

A.5 Annotators Original Coding Manual

Mexico Disappearances Database: Coahuila and Nuevo León

Coding Manual

The objective of this manual is to standardize the ways in which the coding team records cases of disappearances, as represented in major newspaper and media sources in the Mexican states of Coahuila and Nuevo León. One will find in this document instructions for how to respond to each question and the criteria that each coder should use when making entry decisions.

Evento

Q1. Nombre codificadora/or - Name of Coder

First and last name of the coder

Q2. Nombre del Evento (Comenzar con “Desaparición de”)

- Name of Event (Begin with “Desaparición de”) Unless there is an official name for the event (such as Ayotzinapa 34), the event is named after the individual disappeared. Example: “Desaparición de José González”

Q3. Fuente Periódico - Newspaper

Official name of the newspaper with the Mexican state in parentheses. Example: El Norte (Nuevo León)

Q4. Artículo(s) de la Noticia (Citación del Estilo de Chicago) - Article(s) of the Newspaper (Chicago Style Citation)

Cite in Chicago Style all articles you will use to answer the questions. For a brief guideline for how to cite in Chicago Manual of Style (16th Edition), visit the Purdue Online Writing Lab’s Chicago Style webpage: <https://owl.english.purdue.edu/owl/resource/717/01/>.

Víctima

Q5. Nombre de Pila - Name of Victim

Enter the full name of the victim

Q6. Alias - Alias

Enter the known alias of the victim, if it exists.

Q7. Sexo - Sex

Enter the sex (male - female binary) of the victim.

Q8. Edad al momento de la desaparición - Age at the time of disappearance

Enter the age of the victim, if it is given or if you can figure it out from the information in the article.

Q9. Lugar de nacimiento - Place of birth

Enter the place of birth (format: City, State) of the victim, if it is given.

Q10. Nacionalidad - Nationality

Select or enter the nationality of the victim. If “other,” specify in the Spanish feminine form (“otra”).

Q11. Situación migratoria al momento de la desaparición - Migratory status at the time of disappearance

Select or enter the migratory/citizenship status of the victim.

Q12. Domicilio habitual - Residence

Enter the residency (format: City, State) of the victim.

Q13. Escolaridad - Education

Enter the highest level of education of the victim, regardless of whether he or she holds a degree at that level. Example: If the victim were in his or her second year of the equivalent of high school (but had not graduated), one would still select “Secundaria - Secondary.” For university studies, select “other” and enter “Universidad,” followed by the subject. Example: Other: Universidad: Filosofía

Q14. Ocupación - Occupation

Enter the occupation of the victim, if it is given.

Q15. ¿La víctima tiene familiar(es) inmediato(s)? Does the victim have (an) immediate family member(s)?

Select whether the victim has an immediate family members and list all that are known. If so, enter the number. Immediate family members include: parents, siblings, children, and spouse. Example: 3 (Madre, Padre, Hermano)

Q16. Orientación sexual de la víctima - Sexual orientation of the victim

Enter the sexual orientation of the victim, if it is given.

Q17. ¿La víctima pertenece a algún pueblo indígena? - Is the victim a member of an indigenous population?

Select whether the victim belongs to a particular indigenous group. If so, enter the name.

Q18. ¿Pertenece la víctima a algún grupo social distintivo? - Is the victim a member of a distinct social group?

Select the particular social group to which the victim belongs, if it is given.

Q19. Especificar grupo/pertenencia - Specify group

If a particular social group is given, specify the group name. Example: PAN, Catolicismo, etc.

Q20. Otros rasgos físicos relevantes de la víctima - Other relevant characteristics of the victim
List any other physical characteristics of the victim you deem relevant and/or important.

Q21. ¿Aparece en el RNPED? - Does the victim appear in the RNPED?

Search whether the individual is listed in the National Registry, RNPED: <https://rnped.segob.gob.mx/>. We suggest entering the gender of the victim in RNPED and an age range of 0-90, and the corresponding state(s). Search in both tabs, “Fuero Federal” and “Fuero Común”

Q22. Víctima: Otra información importante - Victim: Other important information

List any other information about the victim you deem important for the case that is not captured in questions five to 21.

Hechos

Q23. Amenaza(s) previa(s) - Previous threat(s)

Select whether there have been previous threats made against the victim.

Q24. ¿Quién?

Select and/or list who made the previous threats.

Q25. ¿Sobre qué? - About what?

Select and/or enter what is known about the previous threats.

Q26. Método de captura/desaparición - Method of capture/disappearance

Select and/or enter the method through which the victim had been captured, if it is given.

Q27. Fecha de captura:

Enter the date of capture of the victim, if given. If only the month and year is known, leave the day blank. If date of capture is unknown, enter the date of last seen.

Q28. Localización de desaparición o lugar de fue visto por última vez - Location of disappearance or place where they were last seen

Enter the location (format: City, State) of the location of disappearance, if given.

Q29. Tipo de lugar de desaparición - Type of place of disappearance

Select the type of place of disappearance of the victim, if given. Examples of public locations include generally state-run areas: streets, parks, airports, etc. Examples of private locations include places owned by individuals or businesses: homes, restaurants, stores, etc.

Q30. Medio(s) de captura/desaparición. Arma(s) - Mean(s) of capture/disappearance. Arm(s)

Select and/or enter the means of capture, with regards to the (suspected) perpetrator being armed, if given.

Q31. Medios de captura/desaparición. Vehículos - Means of capture/disappearance. Vehicles

Select and/or enter the means of capture, with regards to the type of vehicle used by the (suspected) perpetrator. Examples of official vehicles are police vehicles, military vehicles, etc. Private vehicles are vehicles belonging to a private person or entity.

Q32. Traspasos - Transfers

Detail whether, who, what, where, when, and why there had been transfers of the victim from one place to another or one vehicle to another, if given.

Q33. Cautiverio - Captivity

Select whether the victim had been held in captivity, if given.

Q34. Testigo(s) - Witness(es)

Select whether there had been witnesses to the disappearance, if given.

Q35. Relación del/de los testigo(s) con la víctima - Relation of the witness(es) to the victim

Select the nature of the relationship of the witness(es), as it/they relate(s) to the victim.

Q36. Desenlace - Outcome

Select and/or enter what the outcome is of the disappearance, if it is given.

Q37. Fecha de desenlace - Date of outcome

Enter the date of the outcome, if given. If only the month and year are known, leave the day blank.

Q38. Localización desenlace - Location of outcome

Enter the location of the outcome (format: City, State), if given.

Q39. Tipo de lugar de desenlace (nombre, página 152 del Tesauro de HURIDOCS) - Type of place of outcome (name, page 152 of the HURIDOCS Thesaurus)

Enter the type of place of the outcome (number), as given on page 152 of the Spanish HURIDOCS thesaurus.

Q40. Hechos: Otra información importante - Facts: Other important information

List any other information about the facts you deem important for the case that is not captured in questions 23 to 39.

Perpetradores

Q 41. Perpetradores. Cantidad - Perpetrators. Quantity

Select the number of perpetrators, if given.

Q42. Categoría de probable perpetrador - Category of probable perpetrator

Select the type(s) of probable perpetrators, if given. Even if this information is alleged and not confirmed, please report it. Click all categories that are reported in the narrative as probable perpetrators, (e.g. if armed perpetrators were wearing state police vests, but unsure if they were actually police, select Policía Estatal + Grupos o individuos vinculados al crimen organizado)

Q43. Relación con la víctima - Relation to the victim

Select and/or enter the relationship between that the perpetrator(s) has/have to the victim, if given.

Q44. Involucramiento del perpetrador - Involvement of the perpetrator

Select and/or enter the ways in which the perpetrator(s) was/were involved in the disappearance, if given.

Q45. Perpetradores: Otra información importante - Perpetrators: Other important information

List any other information about the perpetrators you deem important for the case that is not captured in questions 41 to 44.

Procedimiento Judicial

Q46. Contactos oficiales. Cantidad - Official contacts. Quantity

Enter the number of contacts that the family, and/or friend(s), and/or representative(s), and/or organization(s) has/have had with officials, if given. Officials include police, prosecutors, etc.

Q47. Persona(s) que hicieron el/los contacto(s) - Person(s) who made the contact(s)

Select the type of people who have made contacts, if given.

Q48. Contactos: ¿Con quién? ¿Qué respuestas? - Contacts: With whom? Which responses?

Select the type of responses associated with each person contacted. If “no information,” select the cross row-column “sin información - sin información.”

Q49. En caso de seleccionar "Other(s)" de las hileras a la izquierda, especificar - In the case of selecting "Other" from the rows on the left, specify.

Specify the “Otra(s)” from the previous question (48).

Q50. ¿Se presentó el caso a una instancia de activación de búsqueda de personas? - Was a search for the victim activated?

Select whether an official victim search has been activated, if given.

Q51. Fecha de activación del mecanismo - Date of mechanism activated

Select the date when the victim search was activated, if given. If only the month and year are given, leave the day blank.

Q52. ¿Interpuso una denuncia penal? Has a penal complaint been lodged?

Select whether a penal complaint has been lodged against the (suspected) perpetrator(s).

Q53. Estatus de las órdenes de aprehensión - Status of the arrest warrant

Select the status of the arrest warrant against the (suspected) perpetrator(s), if given.

Q54. Características de el/los detenido(s) - Characteristics of the detained individual(s)

Select all the appropriate characteristics that apply to the detained individual(s)

Q55. Se ha iniciado un proceso judicial? - Has a court case been initiated?

Select whether a court case has been initiated, if given.

Q56. Tipo de tribunal en el que se abrió el expediente judicial - Type of court in which the case file was opened

In reference to question 55, select what type of court it is in which the case file is opened, if given.

Q57. Sentencia - Sentence

Select whether a sentence has been given against the perpetrator(s)/detained individual(s) against whom the court case was, if given.

Q58. Tipo de sentencia - Type of sentence

Select all appropriate types of sentences given against the perpetrator(s)/detained individual(s) against whom the court case was, if given.

Q59. Procedimiento Judicial: Otra información importante - Judicial procedure: Other important information

List any other information about the judicial procedure you deem important for the case that is not captured in questions 46 to 58.

Observaciones y Comentarios

Q60. Observaciones y comentarios - Observations and comments

Enter any other observations or comments that you deem important for the case, as a whole, that may not fit into questions 22, 40, 45, and/or 59.

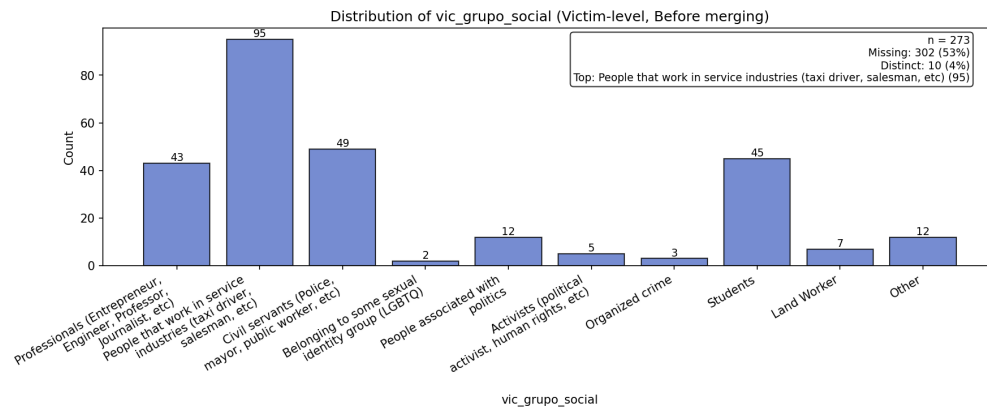


Figure A.6: Distribution of vic grupo social categories per victim before merging.